

IPL Auction Scoring System

Technical Specification (Final Version)

Project

IPL Mock Auction System

Purpose

This document defines the complete scoring model used to determine the winner of the IPL Mock Auction event.

It specifies:

- required data inputs
- scoring formula
- squad validation rules
- calculation steps
- tie-break logic

This document is intended for backend implementation.

1. Auction Context

Each team participates in an IPL-style player auction.

Auction parameters:

Teams per auction: 10

Players per team: 15

Total purse: 120 Cr

Player pool: A, B, C grade players only

D grade players excluded from auction

Player base prices:

```
A grade → 2.5 Cr
```

```
B grade → 1.5 Cr
```

```
C grade → 0.5 Cr
```

2. Squad Constraints (Validation Rules)

Each team must satisfy these conditions after the auction.

Total players:

```
15 players
```

Minimum role requirements:

```
BAT ≥ 3
```

```
BOWL ≥ 3
```

```
AR ≥ 2
```

```
WK ≥ 2
```

Overseas players:

```
2 ≤ overseas ≤ 5
```

If a team violates any rule:

```
TeamStatus = ELIMINATED
```

```
FinalScore = 0
```

Eliminated teams are excluded from winner calculation.

3. Final Score Formula

The winner is determined by the FinalScore.

```
FinalScore =
```

```
BaseScore  
  
+ CaptainBonus  
  
+ ViceCaptainBonus  
  
+ SquadBalanceBonus  
  
+ EfficiencyBonus  
  
+ BrandBonus
```

4. BaseScore

BaseScore measures the strength of the selected playing XI.

Only Top 11 players are considered.

Role weights are applied to slightly reduce bowler dominance.

Role weights:

BAT → 1.00

AR → 1.00

WK → 1.00

BOWL → 0.96

Formula:

```
BaseScore =  
  
 $\sum (\text{role\_weight} \times \text{rating}^{1.15})$   
  
for all players in Top11
```

Where:

```
rating = player rating from dataset
```

5. Captain Bonus

Captain contributes double value.

Since the captain's rating is already counted once in BaseScore, we add one more term.

```
CaptainBonus = rating_captain^1.15
```

Total captain contribution:

```
2 × rating^1.15
```

6. Vice Captain Bonus

Vice captain contributes 1.5× value.

```
ViceCaptainBonus =
```

```
0.5 × rating_vc^1.15
```

Total vice captain contribution:

```
1.5 × rating^1.15
```

7. Squad Balance Bonus

Balance bonus rewards teams with well-structured 15-player squads.

Ideal squad composition:

```
BAT = 5
```

```
BOWL = 5
```

```
AR = 3
```

```
WK = 2
```

```
OVERSEAS = 3
```

Penalty calculation:

```
Penalty =
```

```
|BAT - 5|
```

```
+ |BOWL - 5|
```

```
+ |AR - 3|
```

```
+ |WK - 2|
```

```
+ |OVERSEAS - 3|
```

Balance bonus:

```
SquadBalanceBonus =
```

```
max(0 , 30 - Penalty × 4)
```

Range:

```
0 - 30 points
```

8. Efficiency Bonus

Efficiency rewards teams that spend their purse close to the full budget.

Definitions:

```
TotalSpent = 120 - purse_remaining
```

Efficiency formula:

```
EfficiencyBonus =
```

```
max(0 , 15 - |TotalSpent - 110| × 0.3)
```

Maximum:

15 points

Interpretation:

Teams spending close to 110 Cr receive the highest bonus.

9. Brand Bonus

Brand bonus adds a small advantage based on franchise brand value.

Normalization:

```
NormalizedBrand =  
  
team.brand_score / max(all brand_scores)
```

Brand bonus:

```
BrandBonus =  
  
NormalizedBrand × 5
```

Range:

```
0 - 5 points
```

10. Required Data Inputs

The scoring engine requires the following attributes.

Player Attributes

```
player_id  
  
player_name  
  
rating  
  
category (BAT/BOWL/AR/WK)
```

nationality

grade

Team Attributes

team_id

purse_remaining

brand_score

User Selections

Top11 players

Captain

Vice Captain

11. Score Calculation Workflow

Backend scoring process:

Step 1 — Validate Squad

Check:

squad size

role minimums

overseas limits

If invalid:

team eliminated

Step 2 — Select Top 11

Teams choose:

```
11 players from squad
```

Step 3 — Assign Captain & Vice Captain

Selected from Top 11.

Step 4 — Compute BaseScore

Apply:

```
rating^1.15
```

```
role weights
```

Step 5 — Compute Bonuses

Calculate:

```
CaptainBonus
```

```
ViceCaptainBonus
```

```
SquadBalanceBonus
```

```
EfficiencyBonus
```

```
BrandBonus
```

Step 6 — Compute FinalScore

```
FinalScore =
```

```
BaseScore
```

```
+ CaptainBonus
```

```
+ ViceCaptainBonus

+ SquadBalanceBonus

+ EfficiencyBonus

+ BrandBonus
```

12. Winner Determination

The team with the highest FinalScore wins.

13. Tie-Break Rules

If FinalScore is tied:

Priority order:

1. Higher BaseScore
2. Higher SquadBalanceBonus
3. Higher purse_remaining
4. Higher captain rating

14. Implementation Notes

Important backend considerations:

- All calculations should use floating-point precision
- Ratings exponentiation uses:

```
rating^1.15
```

- Squad balance must use the full 15-player squad
 - BaseScore uses only Top 11 players
 - Captain and vice captain must belong to Top 11
-

15. Summary

The scoring model rewards:

- strong players
- balanced squads
- smart spending
- strategic captain choices

The system has been validated through 60 simulated auctions and has shown stable competitive outcomes.